

# KHALED ABDELHADI

Thermal & Fluid Engineer for vehicle cooling, hydraulics and heat transfer.

Mechanical Engineer (M.Eng.) · 1D-CFD · Hydraulics · CAD & FEM

— Comfortable outside my core field — quick to learn, quick to contribute.

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German / Egyptian citizen

German Professional (B2) · English Fluent · Arabic Native

5

## YEARS

Thermal & Fluid Simulation

3

## DRIVETRAINS

Battery-Electric, Diesel & Hydrogen Fuel Cell

2

## DEGREES

B.Sc. & M.Eng. in Mechanical Engineering

3

## LANGUAGES

German, English, Arabic

## PROFILE

Mechanical engineer (M.Eng.) with **five years in thermal management, hydraulics and simulation** at MAN Truck & Bus and TRATON R&D. I model cooling systems, calculate pressure loss and pipe sizing, and validate everything against CFD and test data. The results become **design decisions**: component sizing, pipe routing, pump operating points and failure prevention. Hands-on background in 3D design, FEM optimization, composites and mold manufacturing.

Thermal Management

1D CFD / GT-SUITE

Hydraulics

Battery & Fuel-Cell Cooling

Pressure-Loss & Pipe Sizing

FEM / FEA

Stress & Stiffness

Structural Optimization

3D CAD Design

2D Drawings & Tolerances

Failure Analysis

Test Correlation

Manufacturing

Python · MATLAB

## EXPERTISE

### • Thermal

Thermal management, heat transfer, thermodynamics, cooling circuits (battery, e-motor, HV, cabin, refrigerant, hydrogen fuel cell), heat exchangers, pipe insulation, immersion cooling, transient drive-cycle simulation, extreme-ambient, worst-case and safety (FTTI) scenarios, thermal hotspot analysis

### • Simulation

1D CFD / system simulation (GT-SUITE, expert), 3D CFD correlation, FEM / FEA structural simulation (Altair HyperWorks; Abaqus / Nastran basic), electromagnetic simulation (ANSYS Maxwell), plastic-flow simulation, model calibration and validation against test data, simulation methodology

### • Manufacturing

Injection mold (steel tool) design, CNC programming, composite layup and curing, prototype build, supplier data evaluation, molding-defect elimination, manufacturing documentation

### • Hydraulic & Fluid

Hydraulic circuit design and modelling, pressure loss and pipe sizing calculations, flow rate and flow distribution analysis across interconnected circuits, pump sizing and operating points, hydraulic control elements (2-way and 3-way valves, check valves, thermostats, flow restrictors), venting and elevation design, hydraulic validation against measurements, P&ID reading, industrial piping (ASME B31.1 / B31.3 / B36, single- and two-phase pipe hydraulics, valves, flanges)

### • CAD & FEM

3D part and assembly design, 2D drawings with tolerances (DIN / ISO), mechanical design of components, stress and stiffness assessment, structural optimization (weight, torsional stiffness), CFRP composites, BOM creation; CAD: PTC Creo, SpaceClaim, Revit (BIM)

### • Tools & Methods

MATLAB/Simulink, Python (data analysis, automation), AUDI loss-map toolbox, JIRA, Agile / Scrum, V-model / ASPICE, failure analysis and root-cause investigation, trade-off studies, KPI tracking, technical documentation

## EDUCATION

M.Eng. Design & Development in Mechanical & Automotive Engineering

B.Sc. Mechanical Engineering

Hochschule Esslingen, Germany · 2019 – 2021

Alfaisal University, Riyadh, Saudi Arabia · 2013 – 2018

## EXPERIENCE

### 1D Simulation Expert, Vehicle Cooling Systems

TRATON R&D Germany GmbH, Nuremberg, Germany

Apr 2025 – Feb 2026

11 months

### 1D Simulation Engineer, Vehicle Cooling Systems

MAN Truck & Bus SE, Nuremberg, Germany

May 2022 – Apr 2025

3 years

GT-SUITE • 3D CFD CORRELATION • TEST-BENCH DATA • JIRA / AGILE

- Led thermal-management simulation for electric and diesel truck cooling systems; owner of the 1D cooling-circuit models used for concept decisions and release.
- Built 1D hydraulic and thermal models from 3D pipe geometry and supplier data; calibrated coolers and circuits against 3D CFD and physical measurements with strong correlation.
- Performed pressure loss, flow rate, flow distribution and pipe sizing calculations across interconnected battery, HV-component, cabin and refrigerant circuits; sized pumps, operating points and hydraulic control elements (2-way and 3-way valves, check valves, restrictors), improving energy efficiency.
- Modelled the refrigerant circuit together with the connected coolant circuits as one system, optimizing their interaction for lower power consumption.
- Ran worst-case, extreme-ambient and safety scenarios (incl. FTTI) and derived design changes: venting positions, flow restrictors, pipe diameters and elevations.
- Covered the cooling side of hydrogen fuel-cell drivetrains alongside the battery-electric and diesel platforms; included pipe insulation in coolant-line models and assessed immersion cooling for a customer-specific project.
- Performed failure analysis on cooling issues and supported component selection with simulation-based trade-off studies for suppliers and design teams.
- Created configurable variant sub-models (e.g. battery count) reused across vehicle platforms; wrote the team's simulation methodology, cutting onboarding time and standardizing model quality.

### 1D/2D Simulation Engineer, E-Motor Cooling

MAN Truck & Bus SE, Nuremberg, Germany

Dec 2021 – May 2022

6 months

ANSYS MAXWELL • AUDI LOSS-MAP TOOLBOX • GT-SUITE

- Simulated e-motor electromagnetic behaviour and losses; validated torque/speed against measurements and generated loss maps for the thermal model.
- Calibrated the GT thermal model to sensor data, achieving temperature predictions within measurement tolerance.
- Identified thermal hotspots, demagnetization risk and previously unaccounted system losses before hardware testing.

### 1D Simulation Student, E-Motor Cooling

MAN Truck & Bus SE, Nuremberg, Germany

Oct 2020 – Mar 2021

6 months

- Built a bus e-motor 1D thermal model in GT-SUITE from scratch, including oil-circuit design with conduction and convection paths; became the base model for all later e-motor cooling studies.
- Converted 3D geometry to 1D and validated transient thermal simulation against drive-cycle test data.

### Structural Simulation Engineer, CFRP Monocoque (Formula Student)

Esslingen Rennstall, Esslingen, Germany

Mar 2020 – Aug 2020

6 months

ALTAIR HYPERWORKS • FEM • CFRP

- Optimized the CFRP monocoque of the Stallardo 19 race car with FEM for lower weight and higher torsional stiffness against structural requirements.
- Delivered the production-ready BOM and 2D schematics for the Stallardo 20 chassis, translating simulation results into manufacturing documentation.

### Engineering Development & Management Intern

A123 Systems GmbH, Stuttgart, Germany

Jan 2020 – Aug 2020

8 months

- Built project KPIs from JIRA tickets across V-model (ASPICE) phases, enabling data-driven milestone forecasting and improving team velocity.
- Analysed BMS hardware/software architecture, mechanical battery design and safety-testing protocols.

### Mold Design & Manufacturing Intern

Alfanar Electric, Riyadh, Saudi Arabia

Jul 2017 – Jan 2018

7 months

PTC CREO • CNC • PLASTIC-FLOW SIMULATION

- Designed steel injection molds as 3D assemblies with 2D drawings, programmed CNC machining and ran plastic-flow simulations to identify and eliminate molding defects.

### CFRP Design & Manufacturing Member

Shell Eco-Marathon Asia, Singapore

Jun 2016 – Jun 2018

2 years

- Designed the vehicle exterior in Creo, built CNC-machined foam molds, led carbon-fibre layup and curing; monocoque validated through tensile testing.

## CERTIFICATIONS

### Design of Industrial Piping Systems

L&T EduTech / Coursera · pipe hydraulics, ASME B31 · [verify](#)

2026

### AI for Design and Optimization

University of Michigan / Coursera · [verify](#)

2026

### Python for Data Science, AI & Development

IBM / Coursera · [verify](#)

2026

### Computational Fluid Mechanics

Coursera project · airflow around a spoiler · [verify](#)

2026

### BIM Fundamentals for Engineers

L&T EduTech / Coursera · [verify](#)

2026

### Claude Code 101

Claude Academy (Anthropic) · completion badge · [verify](#)

2026

### SimOps Fundamentals

Simulation process automation

2025